

Parth Srivastava

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LinkedIn

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Skills

Programming: Python, SQL, TypeScript/JavaScript, KQL (Kusto Query Language)

AI & LLM Systems: AI Agents, Recursive Planning, RAG, LLM Fine-Tuning, PEFT, Synthetic Data Generation, NLP

Frameworks & Backend: FastAPI, PyTorch, LangChain, LangGraph, Hugging Face, SQLAlchemy, Pydantic, Next.js, Scikit-learn

Databases & Security: PostgreSQL, MongoDB Atlas, SQLite, AES-256-GCM Cryptography, Clerk / JWT Authentication

Cloud, DevOps & Tools: AWS, Docker, vLLM, RunPod, Novita AI, Vast.ai, Linux, Apache Airflow, Prometheus, Locust, Git, GitHub

Experience

AI/ML Engineer, 4Way Technologies

Nov 2025 – Present

- Deployed 20+ AI models on RunPod and Novita using Docker, vLLM, and FastAPI.
- Built Apache Airflow ETL pipelines with Bronze, Silver, and Gold data lake architecture, processing 10K+ records daily.
- Built a RAG-based chatbot with real-time knowledge synchronization, supporting dynamic document ingestion, updates, and deletion.
- Developed a multilingual AI assistant integrated with RAG for context-aware conversations across multiple languages.
- Performed API load testing using Locust and monitored latency, throughput (RPS), error rates, and concurrent user performance with Prometheus.
- Developed image generation workflows using Stable Diffusion, ComfyUI, and Automatic1111, serving 200+ users.

Projects

Nodeva — Multi-Cloud GPU Orchestrator

Modeva

Python, FastAPI, PostgreSQL, MongoDB Atlas, Docker, SQLAlchemy, LangChain, Groq, SSE, AES-256-GCM, PyJWT

- Architected Modeva, a unified multi-cloud GPU platform for comparing live spot prices, checking GPU availability, and deploying compute instances across RunPod, Novita AI, and other providers.
- Engineered end-to-end pod management APIs for provisioning, starting, stopping, and deleting instances, along with creating and attaching persistent network storage volumes across global datacenters.
- Built an autonomous natural-language assistant using recursive dependency planning and real-time streaming (SSE) that finds the best spot rates and launches instances from simple text prompts.
- Secured the platform with AES-256-GCM encryption to safely store provider API keys, Clerk JWT authentication, and 1-click interactive approval cards to verify parameters and costs before launch.

LLM Fine-Tuning Pipeline

Source Code

Python, Hugging Face Transformers, PEFT, PyTorch, QLoRA, RunPod, vLLM

- Fine-tuned Qwen2.5-0.5B using Full Fine-Tuning and QLoRA on RunPod GPUs for structured JSON generation.
- Built a synthetic data generation pipeline to create 10K+ instruction samples for fine-tuning.
- Implemented custom tokenization and label masking, training only on assistant responses by masking prompt tokens (~100) for supervised fine-tuning.
- Organized the dataset into multiple progressive stages (P0, P1, ...), iteratively removing conflicting samples to improve data consistency and instruction quality.

Awards & Achievements

Student Leader, SIH 2025

2025

- Led the team in Smart India Hackathon 2025, contributing to solution design and technical decision-making.

Education

Raj Kumar Goel Institute of Technology

B.Tech in Computer Science & Engg. (Data Science) — CGPA: 7.4

Ghaziabad, UP

Aug 2022 – Sep 2026